Income classification and consumer clustering project report

Yang Yang

March 2026

1 Problem and result summary

1.1 Income level classification

We developed predictive models to classify individuals into two income groups: those earing greater than $50,000$ annually and those earning less than $50,000$ annually. The classification task is motivated by a retail business client to better target marketing efforts toward different consumer groups. We trained a baseline logistic regression model and an XGBoost model. The XGBoost model achieved significantly better performance than the baseline model with the following detailed metrics.

- ROC-AUC: 0.9531
- F-1:0.6221
- Precision-Recall AUC: 0.6796.

Note that the model can be modified to incrase the classification accuracy of either class at the cost of classification accuracy of the other class.

1.2 Segmentation model

We developed an unsupervised clustering model that can be used to group customers by their demographic, employment, financial information. This infers our rectail client potential consumer behavior differences between groups and help them make targeted marketing effort. Using a combination of PCA dimension reduction and K-Means clustering, we found three distinct groups of customers and gave some ideas for product and marketing recommendations:

- immigrant working professional: specialty (imported) goods, communication services
- non-immigrant working professional: groceries, appliances, US-culture-based products
- children, veterans, seniors: toys, casual sport goods, entertainment.

To reach these populations effectively, one can consider digital platform (e.g. social media) for group 1 and 2, and TV commercial for group 3.

2 Classification model

2.1 Exploratory data analysis

2.1.1 Feature summary

For convenience, we convert the target label to binary values by setting the value $<$ \$50000 to 0 (negative) and $>$ \$50000 to 1 (positive).

- total number of data points: 199523 (196294 after removing 3229 duplicates)
- numerical data:
  - profession-related: 'detailed industry recode', 'detailed occupation recode', 'wage per hour', 'member of a labor union', 'num persons worked for employer', 'own business or self employed', 'weeks worked in year',
  - personal information: 'age', 'sex', 'weight', 'live in this house 1 year ago', "fill inc questionnaire for veteran's admin", 'veterans benefits',
  - finance-related: 'capital gains', 'capital losses', 'dividends from stocks',
  - survey year
- categorical data:
  - profession-related: 'class of worker', 'major industry code', 'major occupation code', 'reason for unemployment', 'full or part-time employment stat',
  - education-related: 'education', 'enroll in edu inst last wk',
  - personal information:'marital stat', 'race', 'hispanic origin', 'detailed household and family stat', 'detailed household summary in household', 'family members under 18', 'country of birth father', 'country of birth mother', 'country of birth self', 'citizenship', 'tax filer stat',
  - location information: 'region of previous residence', 'state of previous residence', 'migration code-change in msa', 'migration code-change in reg', 'migration code-move within reg', 'migration prev res in sunbelt',
- target: 93.7% (183912) are negative (0) and 6.3% (12382) are positive (1).

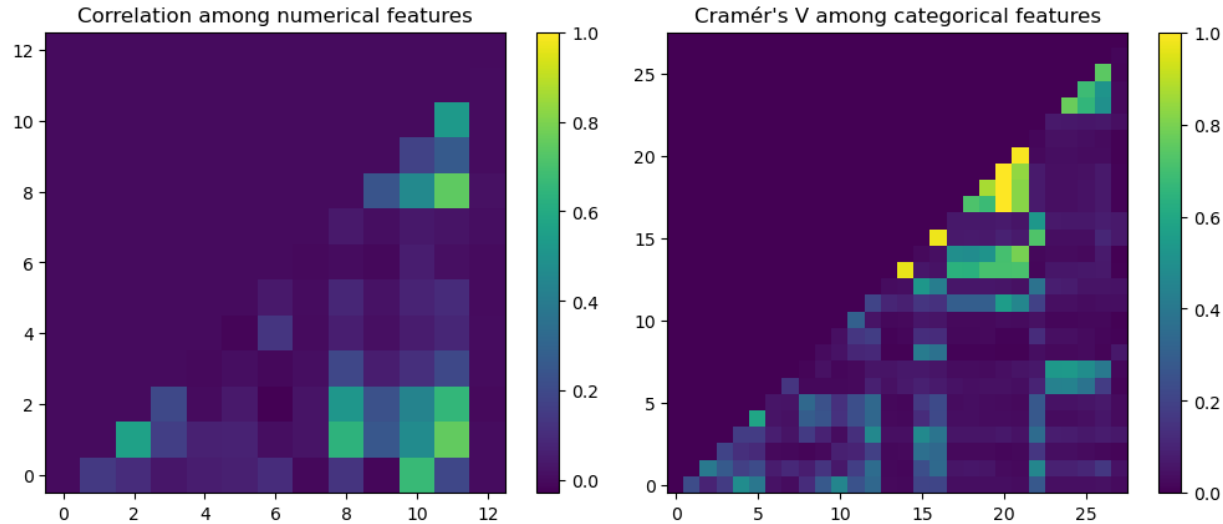
The only feature with missing data is "hispanic origin" where we fill in the missing data with "missing".

2.1.2 Correlations

To investigate dependencies among features, we compute correlations among numerical features and use Cramér's V statistics for categorical features. As shown in Figure 2.1.2, there are no strong correlations between numerical features.

Categorical features on the other hand, exhibit some relatively high Cramér's V values. Upon closer inspection, we found that pairs of categorical features with Cramér's V value

greater than 0.7 often invovle categories that are extremely unbalanced, which may inflate the Cramér's V values. We also evaluated mutual information as an alternative measure of association among categorical features and obtained similar results. Based on this analysis, we decided not to drop any features at this stage. Retaining all features is reasonable for the XGBoost model, as it can inherently handle feature redundancy and is robust to irrelevant or correlated inputs. By keeping all features, we allow the model to leverage subtle interactions and patterns that might otherwise be lost if features were removed prematurely.



2.2 Evaluation metrics

The dataset is highly imbalanced, with over 93% of the samples labeled as negative. Since the client did not specify whether we are prioritizing identifying positive or negative instances, we select the balanced F1 score as our primary evaluation metric. We also report additional metrics including accuracy, precision, recall, and the Precision-Recall AUC (PR AUC).

We first train a baseline logistic regression model due to its simplicity and interpretability. Subsequently, we train an XGBoost model to improve performance over the baseline.

2.3 Baseline logistic regression model

For the baseline, we use a stratified 80-20 train-test split. The model is trained on 80% of the data and evaluated on the remaining 20%. Categorical features are one-hot encoded, while numerical features are standardized using a standard scaler.

The model's performance is summarized in Table 1.

We observe that while overall accuracy is high, the F1 score is relatively low due to the strong class imbalance.

2.4 XGBoost model

For the XGBoost model, we adopt frequency encoding for categorical features to reduce dimensionality compared to one-hot encoding. Each categorical value is mapped to its

ROC AUC	0.9440
F-1	0.5122
PR AUC	0.6073
precision	0.7268
recall	0.3954

Table 1: Baseline logistic regression model performance.

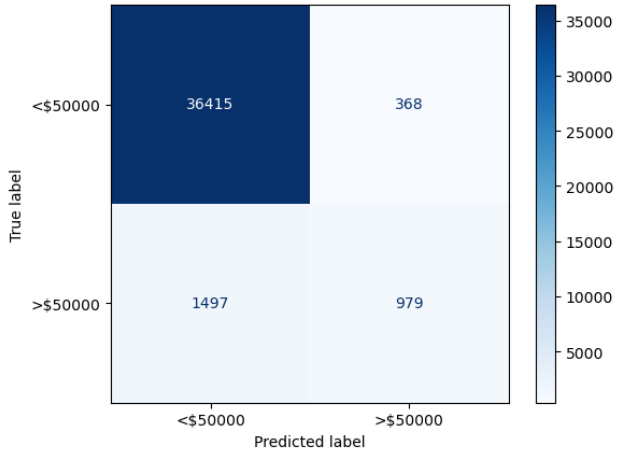


Figure 1: Baseline Logistic regression model confusion matrix.

frequency in the training set. Numerical features are left unscaled, as XGBoost is insensitive to feature scaling.

Parameter	Values
Learning rate	[0.05 0.1 0.2]
Max depth	[4 6 8]
Positive class weight scaling	[2 4 6]
Number of estimators	[100 200 500]

Table 2: XGBoost parameter space.

Hyperparameter tuning is performed via grid search over the parameter space given in Table 3. We train a model for each combination of these four parameters and select the configuration that maximizes the F1 score.

We employ cross-validation to obtain a robust estimate of model performance when training each model. After an initial 80-20 train-test split, the training data is further split into five folds. For each fold, we use it as validation data and train on the remaining four folds. As a result, the XGBoost model with each parameter set is trained five times, and the reported performance is averaged across folds. Since the F1 score is threshold-dependent and the default threshold of 0.5 may be suboptimal for imbalanced data, we optimize F1 over 100 thresholds ranging from 0.05 to 0.95.

Once we find the optimal hyperparameter value combination, we train a model using these hyperparameters on the entire training set and evaluate this model on the test set.

The training scheme is shown in figure 2.

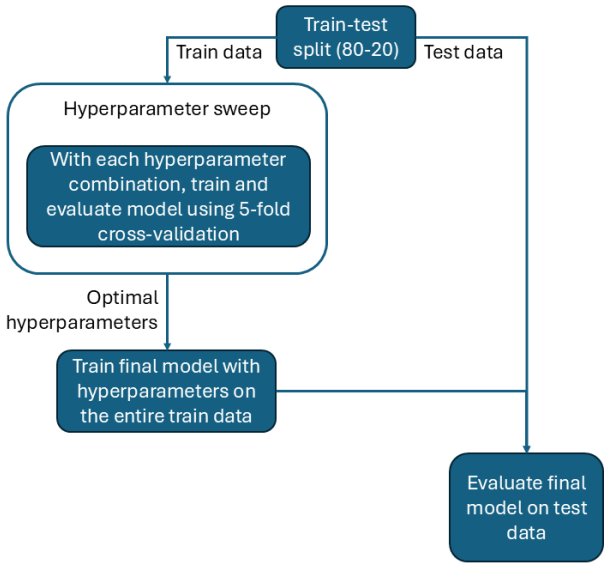


Figure 2: XGBoost training scheme

The optimal hyperparameters are listed in Table 3, and the final model performance is summarized in Table 4.

Parameter	Optimal Value
Learning rate	0.1
Max depth	8
Positive class weight scaling	4
Number of estimators	200

Table 3: Optimal hyperparameters for XGBoost based on F1 score.

Metric	Value
ROC AUC	0.9531
F1 Score	0.6221
PR AUC	0.6796
Precision	0.6091
Recall	0.6357

Table 4: Performance of the XGBoost model on the test set.

While the overall performance improvement is moderate, we observe notable gains in PR AUC and recall. The decrease in precision is an expected trade-off to achieve higher recall, which is often desirable in imbalanced classification tasks.

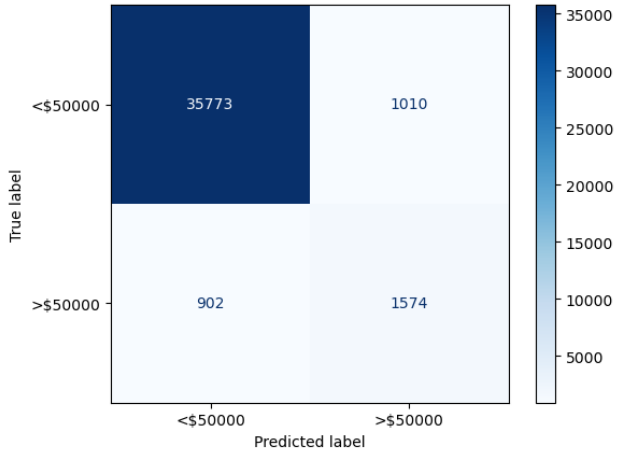


Figure 3: Caption

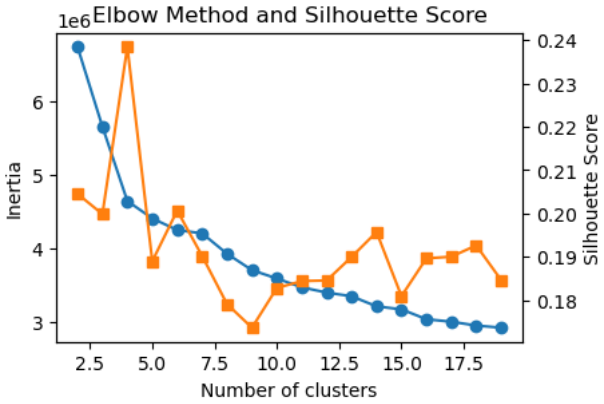


Figure 4: Optimal number of clusters for K-Means.

3 Segmentation model

For the segmentation task, we implement a combination of PCA for dimensionality reduction and K-Means clustering. Since this is an unsupervised portion of the project, the target column is removed from the dataset. Categorical features are first encoded using frequency encoding, and all features are subsequently scaled using a standard scaler. Although the dataset contains only 41 features, training a K-Means model directly on the scaled data resulted in a very low silhouette score (less than 0.2). To address this, we apply PCA, retaining 90% of the variance in the data. The K-Means model is then trained on this dimensionally reduced dataset.

The optimal number of clusters is determined using a combination of the elbow method (based on inertia) and the silhouette score. Based on these criteria, we select $k = 3$ as the number of clusters (Figure 4).

We then analyze the distribution of features across clusters. Figure 5 shows the distribution of numerical features (excluding the survey year) by cluster. The most significant separations between clusters are observed in age, detailed industry and occupation codes,

number of persons worked for employer, veterans benefits, and weeks worked in a year.

- Cluster 1 (87,781 individuals) consists primarily of middle-aged professionals who are immigrants. Only 1% were born in the US, 3% have a birth mother from the US, and 4% have a birth father from the US. Approximately 69% of individuals in this cluster identify as white.
- Cluster 2 (86,392 individuals) represents middle-aged US-born professionals. All individuals were born in the US, with 94% having a birth mother and 93% having a birth father from the United States. About 89% of this cluster identifies as white.
- Cluster 3 (22,121 individuals) appears to comprise mostly children and seniors, as indicated by age, occupation, and veteran benefits. Notably, 75% are categorized as children or armed forces, while the remainder are marked as not in the labor force.

These clusters suggest targeted marketing opportunities for different products:

- Cluster 1: Imported goods, international communication services (e.g., phone plans), and specialty products based on country of origin.
- Cluster 2: Household appliances, groceries, and products related to US holidays and festivals.
- Cluster 3: Toys, home goods, and casual sports or entertainment products suitable for children and retired individuals.

4 Limitation and future work

Due to time constraints, we were unable to perform a comprehensive analysis of the feature space. We observed correlations among features, suggesting that a natural next step would be to combine or reduce features in a meaningful way.

Additionally, we experimented with only a limited set of classification models, hyperparameters, and feature encoding methods. We believe there is considerable potential to improve performance by exploring alternative models, hyperparameter configurations, and combinations of feature engineering techniques.

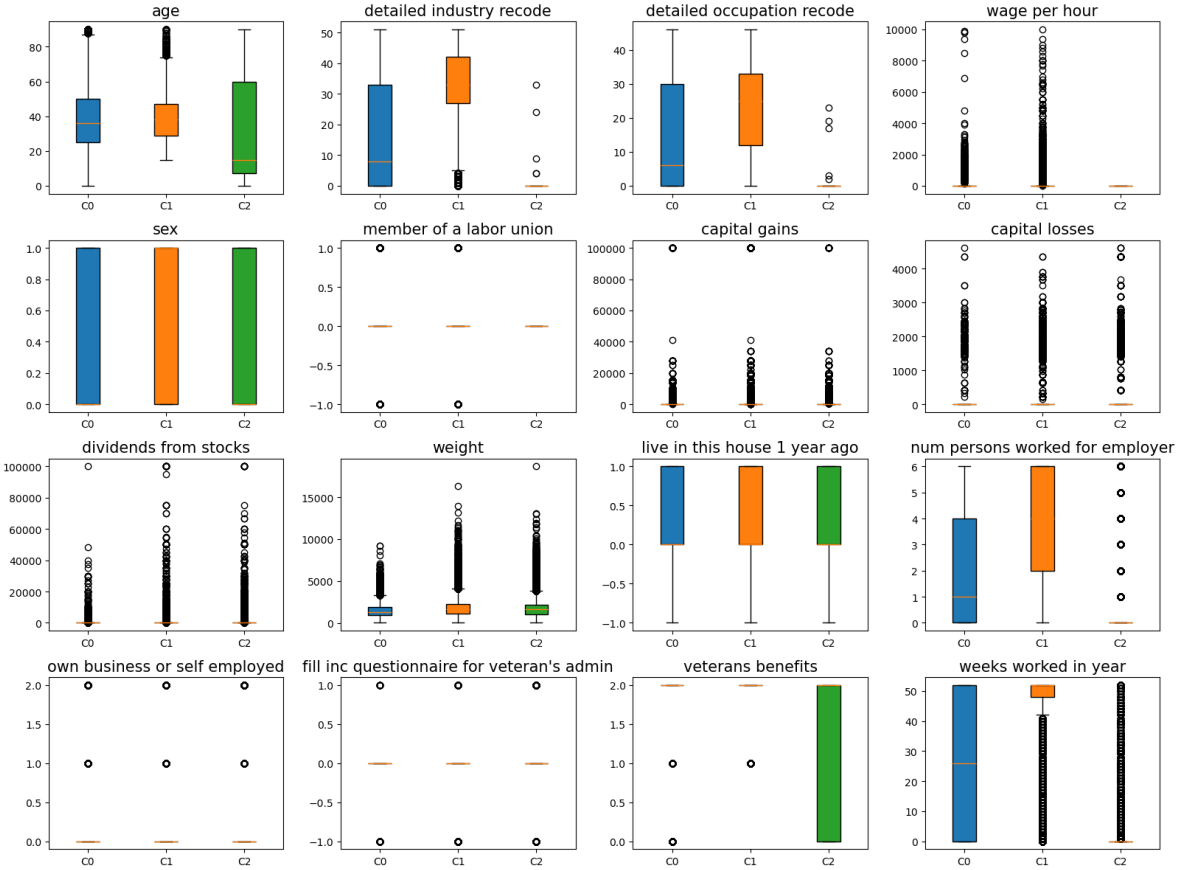


Figure 5: Distribution of numerical features by cluster.